Course 6031D · Semester 6 · Program Elective · Revision 2021

Electric Vehicles

A professional engineering handbook on electric vehicle classifications, EV/HEV drive trains, motor drive technologies, energy storage, charging infrastructure, renewable energy integration, grid support services and EV maintenance.

What this subject is

This course explains the engineering of electric mobility. It starts with the relevance and classification of EVs, then moves into EV drive trains, motors, hybrid operation, regenerative braking, energy storage, charging systems and renewable-energy based charging infrastructure.

Malayalam: EV-□ engine-□□ □□□□ motor, battery, inverter, controller, charger □□□□□□□□□□ □□□□□□ systems. □ subject EV architecture □□□□ charging station □□□ □□□□□□□□□□□□□□□□.

Official syllabus information

Item	Value
Program	Diploma in Electrical & Electronics Engineering / Electrical Engineering & Electric Vehicle Technology / Mechanical Engineering / Tool & Die Engineering / Renewable Energy
Course code	6031D
Course title	Electric Vehicles
Semester	6
Credits	4
Category	Program Elective
Periods	4 per week (L:3, T:1, P:0); 60 per semester

Course objectives

- Express relevance of EVs for future transportation.
- Understand electric motors used in EV applications.
- Summarize energy storage options in EV.
- Identify charging infrastructure requirements.

Course outcomes

- CO1: Outline EV basic concepts and classifications.
- CO2: Identify EV and HEV drive trains.
- CO3: Outline hybridization of energy storage.
- CO4: Explain EV charging integrated with renewable energy.

How to study

Learn EV blocks first: battery, motor, inverter, controller, charger and auxiliary system. Then compare BEV, HEV, PHEV and FCEV. After that study drivetrain, braking, storage, charging station and renewable integration.

Redesigned syllabus roadmap

Official syllabus converted into practical EV engineering learning route.

Module 1 · CO1 · 12 Hours

Basic Concepts of Electric Vehicles and Classifications

Benefits, EV development, general classification, government policies and motor drive technologies.

Introduction and benefits

Electric vehicles use electric propulsion instead of only internal combustion. Energy is stored in battery, fuel cell or hybrid storage and converted to mechanical torque through power electronics and motor drive. Benefits include reduced tail-pipe emission, higher drivetrain efficiency, regenerative braking, lower noise and better integration with renewable energy.

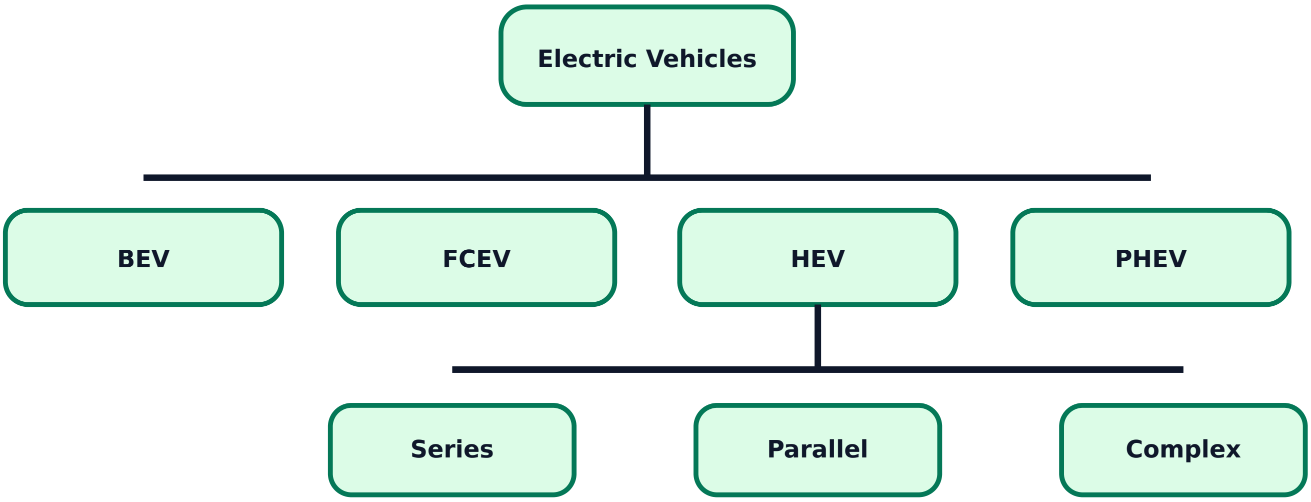
Malayalam: EV-□ fuel burning engine □□□□□□ □□□□. Battery energy inverter □□□ motor-□□□□□□ □□□□□□□□. Braking □□□□□□□ motor generator □□□□ □□□□□□□□□□□□□ energy □□□□□□ battery-□ □□□□□□□□□□.

EV classification

- **BEV:** battery electric vehicle; charged from external supply.
- **FCEV:** fuel cell electric vehicle; fuel cell generates electricity.
- **HEV:** hybrid electric vehicle; engine and motor work together.
- **PHEV:** plug-in hybrid; battery can be charged externally.
- **PE:** pure electric vehicle with electric traction only.

Exam tip: For classification questions, draw a tree diagram and compare source, motor, battery and charging method.

EV classification tree



Government policies and schemes

The syllabus includes FAME-1, FAME-2, transformative mobility and energy storage mission, latest central policies, state policies, subsidies and global EV success stories. These schemes support EV purchase, charging infrastructure, manufacturing and clean mobility adoption.

Motor drive technologies

EV traction motors need high starting torque, wide speed range, high efficiency and reliable control. Major motor drives are DC motor drive, induction motor drive, permanent magnet motor drive and switched reluctance motor drive.

- DC motor: simple control but maintenance due to brushes.
- Induction motor: rugged, brushless, needs inverter control.
- PM motor: high efficiency and power density.
- SRM: rugged rotor, simple construction, torque ripple challenge.

Expected questions

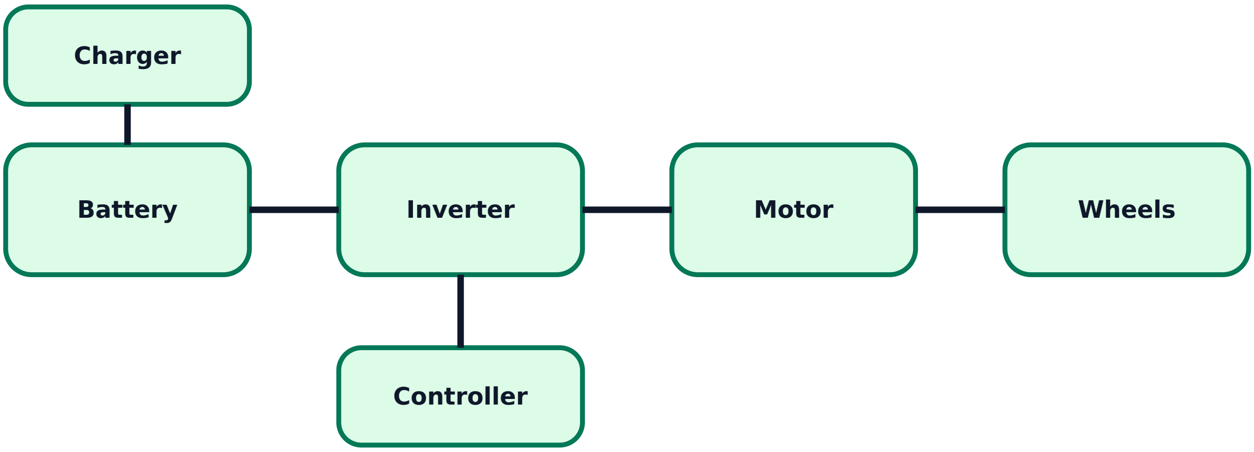
1. Explain benefits of EV over conventional vehicle.
2. Classify EVs and explain BEV, FCEV, HEV and PHEV.
3. Explain FAME schemes and EV policy relevance.
4. Compare DC, induction, PM and switched reluctance motor drives for EV.

Module 2 · CO2 · 14 Hours

EV and HEV Drive Trains

EV subsystems, drive train alternatives, HEV modes, power flow control and braking.

EV subsystem configuration



Main EV subsystems

- Electric propulsion unit: motor, inverter, controller and transmission.
- Energy source subsystem: battery/fuel cell/supercapacitor with management system.
- Auxiliary subsystem: DC-DC converter, 12 V supply, HVAC and control electronics.

Converted EVs modify an existing vehicle platform; purpose-built EVs are designed from the start for battery layout, crash safety, motor placement and thermal management.

Drive train alternatives

Drive train configuration may use single motor with differential, dual motor drive, in-wheel motors or multi-motor systems. Single speed transmission is common because electric motors provide useful torque over a wide speed range.

> **Practical note:** In-wheel motors reduce mechanical transmission parts but increase unsprung mass and need rugged sealing.

HEV modes and power flow

Series HEV uses engine to generate electricity and motor drives wheels. Parallel HEV allows both engine and motor to drive the wheels. Series-parallel combines both. Complex HEV uses advanced power split and multiple power paths. Power flow control decides when power comes from engine, battery or both.

Braking in EV

Regenerative braking converts vehicle kinetic energy into electrical energy and charges the battery. Hybrid braking combines regenerative and mechanical friction braking to maintain safety and braking feel.

Malayalam: Regenerative braking-□ motor generator □□□ □□□□□□□□□□□□□ braking energy battery-□□□□□□ □□□□□□ □□□□□□□□□□□□□. Full stop/urgent brake-□ friction brake □□□□ □□□□.

Expected questions

1. List EV subsystem components.
2. Explain basic EV drive train architecture.
3. Explain series, parallel and series-parallel HEV power flow.
4. Explain regenerative and hybrid braking.

Module 3 · CO3 · 16 Hours

Energy Storage, Battery Charging and EVSE

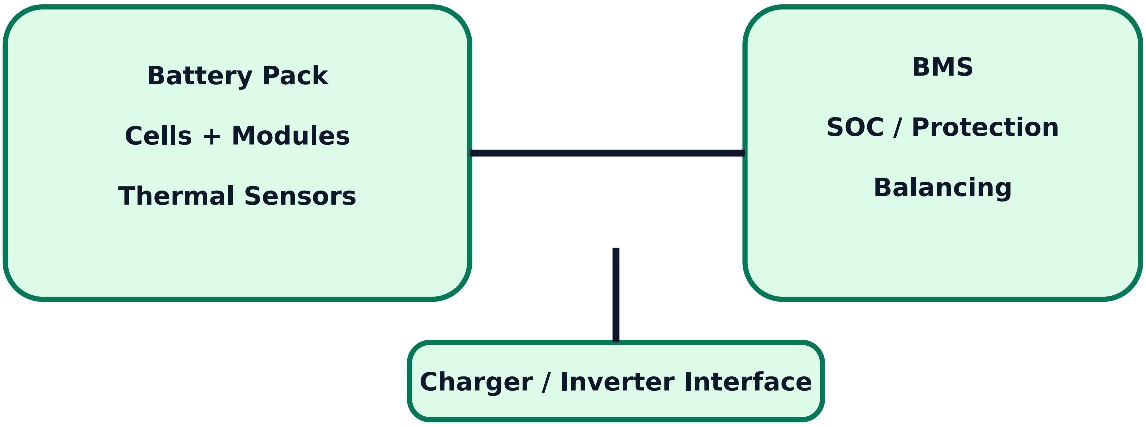
Batteries, ultracapacitors, flywheels, fuel cells, hybrid storage, BMS, power electronics, charging infrastructure and smart charging.

Energy storage technologies

EV storage must provide energy, power, life, safety and acceptable charging time. Syllabus storage devices include batteries, ultra-capacitors, ultra-flywheels and fuel cells. Batteries are described using voltage, capacity, energy, power, SOC, DOD, C-rate, cycle life and efficiency.

- Lead-acid: low cost, heavy, lower energy density.
- Nickel based: better than lead-acid but less common in modern EVs.
- Lithium based: high energy density and common in EVs.
- Supercapacitor: high power, fast charge, lower energy.
- Fuel cell: converts fuel energy to electricity.

Battery and BMS block



Charging technologies

Charging can be domestic, public, fast charging, battery swapping or move-and-charge zone. Charging may use on-board charger or off-board charger. EVSE includes connector, protection, communication, metering and control.

> **Safety note:** EV charging station design must consider earthing, protection, isolation, cable rating, connector standard, communication and emergency shutdown.

Power electronics and drive sizing

Power electronics converters control energy flow between battery, motor, charger and auxiliaries. EV design requires sizing of propulsion motor, inverter, DC-DC converter, storage technology and supporting subsystems.

EVSE and smart charging

EVSE standards and communication protocols enable safe charging and billing. Smart charging controls charging time and power based on grid condition, tariff, renewable energy availability and user requirement.

Expected questions

1. Explain battery parameters used in EV.
2. Compare batteries, supercapacitors, flywheels and fuel cells.
3. Explain BMS functions.
4. Explain on-board/off-board charger and EVSE types.

Module 4 · CO4 · 16 Hours

Charging System Integrated with Renewable Energy

Grid integration, V2X, grid support, renewable charging stations, distributed generation, maintenance and recycling.

EV-grid integration

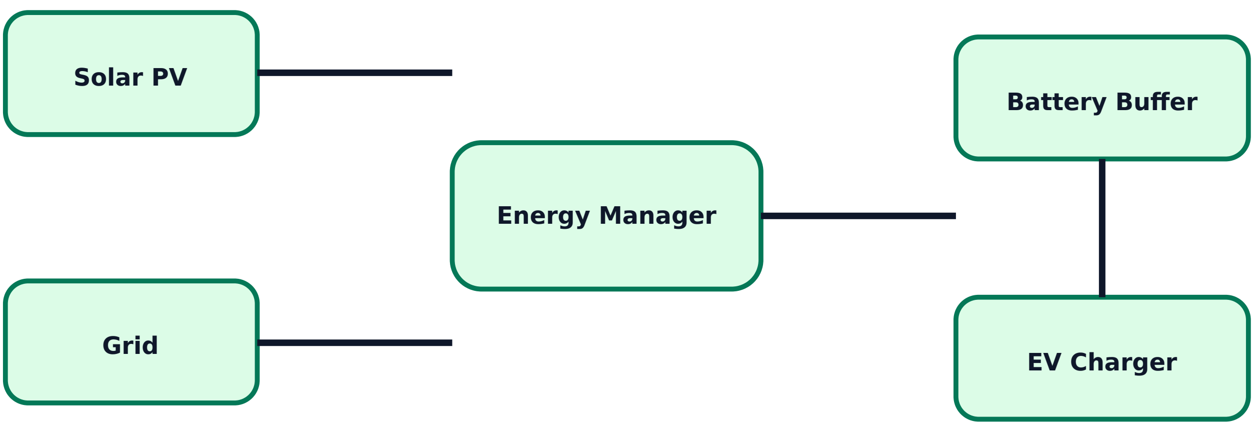
Large EV charging load affects distribution system through congestion, voltage drop, power quality issues and transformer loading. Smart charging, load scheduling and grid upgrades are used for reliable fast charging stations.

V2X and grid support services

V2X means vehicle-to-anything: V2G, V2V, V2H and V2B. EV fleets can provide demand response, voltage/reactive power support and frequency support when aggregated and controlled properly.

Malayalam: V2G-□ EV battery grid-□□ support □□□□□. □□□□□ battery health, user need, charger standard □□□□□□ □□□□□□□□□□□.

Renewable EV charging station



Renewable integration

Renewable energy based charging stations use solar/wind energy, battery storage, grid connection and energy management. Scheduling EV charging during renewable generation improves renewable energy utilization and reduces grid stress.

Maintenance and troubleshooting

EV maintenance includes motor, drivetrain, battery, BMS and charger checks. Faults may include overheating, insulation failure, low battery capacity, charger communication errors, motor controller trips and mechanical drivetrain faults. Battery disposal and recycling are important for environmental safety.

Expected questions

1. Explain impact of EV charging on distribution system.
2. Explain V2X technologies and grid support services.
3. Explain renewable energy based EV charging station.
4. Explain EV battery disposal, recycling and troubleshooting.

Formula / Rule Bank

EV numerical and design rules used in quick problem solving and exam writing.

Battery energy

Energy (Wh) = Voltage (V) x Capacity (Ah)

Used to estimate EV battery size.

Charging time

Charging time = Battery energy / Charger power

Include efficiency and SOC window in real design.

Cost of charging

Cost = Energy consumed (kWh) x Tariff per kWh

Range estimate

Range = Usable battery energy / Energy consumption per km

Regeneration rule

Recovered energy depends on speed, mass, braking event and battery acceptance.

Selection rule

Choose motor and battery by torque, power, voltage, duty cycle, thermal limit, cost and safety.

Charging station rule

Check connected load, transformer capacity, protection, earthing, cable rating and communication.

BMS rule

BMS must monitor voltage, current, temperature, SOC, SOH, balancing and protection.

Solved Examples / Problem Lab

Step-by-step EV calculations and concept applications.

Example 1 - Battery energy

Given: Battery pack voltage = 350 V, capacity = 100 Ah.

Energy = V x Ah = 350 x 100 = 35000 Wh.

Battery energy = 35 kWh.

Example 2 - Charging time

Given: Usable energy to charge = 30 kWh, charger = 7.5 kW.

Time = 30 / 7.5 = 4 h.

Approximate charging time = 4 hours, excluding losses.

Example 3 - Range estimate

Usable battery = 42 kWh. Consumption = 0.16 kWh/km.

Range = 42 / 0.16.

Range = 262.5 km.

Example 4 - Charging cost

Energy consumed = 18 kWh. Tariff = Rs.8/kWh.

Cost = 18 x 8 = Rs.144.

Example 5 - Motor choice

A compact EV needs high efficiency, compact size and good torque control.

Permanent magnet motor is commonly preferred, but cost and magnet availability must be considered.

Practice Section and Expected Questions

Module-wise exam preparation.

One-mark questions

1. Define BEV.
2. What is regenerative braking?
3. Name two EV energy storage devices.
4. What is EVSE?
5. What is V2G?

Short-answer questions

1. List benefits of EV.
2. Compare EV and conventional vehicle.
3. Explain single speed transmission.
4. Explain BMS functions.
5. List EV charging infrastructure types.

Descriptive questions

1. Classify EVs and explain BEV, HEV, PHEV and FCEV.
2. Explain EV drivetrain and HEV power flow modes.
3. Explain energy storage technologies used in EV.
4. Explain renewable energy integrated EV charging station.

MCQ practice

1. In regenerative braking, the traction motor works as: A) heater B) generator C) rectifier D) transformer

2. BMS mainly protects: A) tyres B) battery pack C) chassis D) lights

3. V2G means: A) Vehicle to Grid B) Voltage to Ground C) Vehicle to Gear D) Variable to Grid

Fresh Model Question Paper

Original question paper based on syllabus coverage; not copied from official question papers.

Course 6031D - Electric Vehicles

Time: 3 Hours **Maximum Marks:** 100

Part A - Short Answer

1. Define electric vehicle.
2. What is PHEV?
3. Name any two EV motors.
4. What is BMS?
5. What is V2X?

Part B - Medium Answer

1. Explain benefits of EV including environmental impacts.
2. Compare BEV, HEV, PHEV and FCEV.
3. Explain EV subsystem components.
4. Explain regenerative braking and hybrid braking.
5. Explain EVSE and smart charging.

Part C - Long Answer / Problems

1. Draw and explain general configuration of an electric vehicle.
2. Explain series, parallel, series-parallel and complex HEV power flow.
3. A 400 V, 120 Ah battery is charged by a 12 kW charger. Calculate battery energy and ideal charging time.
4. Explain renewable energy based EV charging station and grid support services.
5. Explain EV maintenance, troubleshooting, battery disposal and recycling.

Complete Model Answers

Answer guide with keywords, diagrams to draw and mistakes to avoid.

Electric vehicle definition

An EV uses electric motor propulsion powered by an energy storage system such as battery, fuel cell or hybrid storage.

EV configuration answer points

Draw battery, BMS, inverter, motor, transmission/wheels, charger, controller and auxiliaries. Explain energy flow from battery to motor and regenerative flow from motor to battery.

Battery problem

Energy = V x Ah = 400 x 120 = 48000 Wh = 48 kWh. Charging time = 48 / 12 = 4 h ideal.

48 kWh and 4 hours ideal charging time.

HEV power flow

Series HEV engine drives generator; motor drives wheels. Parallel HEV engine and motor both can drive wheels. Series-parallel combines both using power split. Complex HEV has advanced multiple power paths.

Renewable charging answer

Include solar/wind source, grid, battery buffer, charger, EVSE, metering and energy management. Mention scheduling, demand response, V2G and grid support.

Common mistakes

Do not write EV as only battery plus motor. Include power electronics, BMS, charger, auxiliaries, thermal management and safety/protection.